

Nikhil Chaudhari

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Education

- 2021 – **Ph.D.** Biomedical Engineering.
University of Southern California, Los Angeles, California, USA.
- 2017 – 2019 **M.S.** Computer Science.
University of Southern California, Los Angeles, California, USA.
- 2013 – 2017 **B.Tech.** Computer Science and Engineering.
Vellore Institute of Technology, Vellore, Tamil Nadu, India.

Research Experience

- 2018 – **Graduate Research Assistant, PhD student**
Aging & Imaging Laboratory, USC.

Local Brain Age. Trained a deep neural network (DNN) using PyTorch to predict biological age of a magnetic resonance imaging (MRI) brain scan to map heterogeneous mapping of aging processes in healthy populations [\[GitHub\]](#) [1].
Brain evolution Generative model. Synthesized high-resolution, participant-specific future (follow-up) MRI using a novel 3D hybrid ControlNet + diffusion model from baseline MRIs of healthy adults, potentially assisting in the prognosis of aging-related brain diseases [27].
Neuroimaging Saliency comparison. Assessed 7 saliency methods for neuroanatomic interpretability in DNNs estimating biological brain age from MRIs, recommending Integrated Gradients as the best approach [10].
Brain age calculation. Trained a deep neural network using TensorFlow to predict biological age of a MRI brain scan [\[GitHub\]](#) [2].
Brain CT segmentation. Developed an automated human head computed tomography (CT) volume segmentation software to calculate white and gray matter volumes for a cross sectional analysis of brain atrophy rates in an atypical population at exceptionally low risk for dementia. Further additions to the software include parcellation of the white and gray matter volumes into more than 160 brain regions [29], [9].
Tractography generation. Developed a pipeline to generate models of white matter fibers (tracts) using diffusion MRIs.
Resting state fMRI connectivity in AD vs. TBI. Computed resting state functional connectivity patterns of Alzheimer's disease (AD), traumatic brain injury (TBI) and cognitively normal participants [18].
Neuroimage cloud computing. Managed AWS S3 and EC2 cloud computing instances for neuroimaging analyses.
System Administrator. Configured, installed, and maintained HPE Linux servers. Installed and configured NVIDIA A100 GPUs for deep learning projects.

Department of Biomedical Engineering, Viterbi School of Engineering, USC.
- 2022 **DICOM Anonymizer.** Developed an open source DICOM (digital medical images) de-identification software. Features include support for customized anonymization through severity levels ('soft', 'medium', 'hard') and selection of individual DICOM tag to anonymize based on user requirements [\[GitHub\]](#).
- 2016 – 2017 **Undergraduate Research Assistant**
School of Computer Science and Engineering, VIT.
Prediction of yoga postures. Developed hardware and a bespoke Android application to track body vitals and predict beneficial postures [21].

Research Experience (continued)

- 2016 ■ *Alteration of Advanced Encryption Standard.* Compared execution times of Advanced Encryption Standard (AES) encryption and decryption based on number of rounds.
Chemistry simulation (Eureka). Simulated chemical reactions at atomic level using basic animations on Java, Oracle 11g and Greenfoot.
- 2015 ■ *Fitness Tracking system using Arduino.* Developed a low-cost pedometer to track calorie expenditure, participant velocity, and ground distance covered.
Detecting and preventing SQLIA. Implemented a demonstration of SQL injection attacks (SQLIA), its detection and prevention.

Employment History

- 2017 ■ **Intern** VirtusaPolaris North Plainfield, New Jersey, USA
Developed a Business Process Management (BPM) application using Pega Rules Process Commander (PRPC) to model, automate, execute, control, measure and optimize business activity flows [19].
- 2014 ■ **Intern** Persistent Systems Nagpur, Maharashtra, India
Calculated user engagement on twitter data using transformations and aggregations with Elasticsearch, Kibana and JSON [40].

Research Publications

Journal Articles

- 1 N. N. Chaudhari, N. F. Chowdhury, O. M. Vega, S. Bhattacharya, and A. Irimia, "Deep learning maps local brain aging in relation to cognition across human adulthood," *Proceedings of the National Academy of Sciences*, 2026.
DOI: 10.1073/pnas.2532233123.
- 2 N. N. Chaudhari, O. M. Vega, P. Imms, *et al.*, "Interpretable deep learning reveals spatiotemporal mri features of brain aging that align with neurodegeneration," *GeroScience*, pp. 1–19, Apr. 2026. DOI: 10.1007/s11357-026-02112-2.
- 3 R. Fox, N. N. Chaudhari, S. Bhattacharya, *et al.*, "Women's reproductive factors predict local brain aging profiles mapped using deep neural networks," *GeroScience*, pp. 1–16, Mar. 2026. DOI: 10.1007/s11357-026-02180-4.
- 4 P. Imms, H. Wang, S. Bhattacharya, *et al.*, "Deep survival modeling to predict future cognitive impairment in unimpaired adults," *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences*, vol. 81, no. 6, glago76, May 2026. DOI: 10.1093/gerona/glag076.
- 5 O. M. Vega, P. Imms, N. N. Chaudhari, W. J. Mack, N. F. Chowdhury, and A. Irimia, "Cognitive performance modulates regional brain age differences in clinical anxiety and depression," *Journal of Affective Disorders*, vol. 413, p. 122 237, 2026, ISSN: 0165-0327. DOI: 10.1016/j.jad.2026.122237.
- 6 X. Xu, N. N. Chaudhari, P. Imms, *et al.*, "Segmentation and morphometry of intracranial internal carotid artery calcification in relation to brain atrophy," *Neuroradiology*, pp. 1–16, Mar. 2026. DOI: 10.1007/s00234-026-03918-9.
- 7 P. Imms, N. N. Chaudhari, D. Cummings, *et al.*, "Physical activity mediates age differences in cognition among tsimane forager-horticulturalists," *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences*, glaf163, 2025, ISSN: 1758-535X. DOI: 10.1093/gerona/glaf163. eprint: <https://academic.oup.com/biomedgerontology/article-pdf/80/8/glaf163/63834859/glaf163.pdf>.

- 8 C. Yin, P. Imms, N. F. Chowdhury, *et al.*, "Deep learning to quantify the pace of brain aging in relation to neurocognitive changes," *Proceedings of the National Academy of Sciences*, vol. 122, no. 10, e2413442122, 2025.  DOI: 10.1073/pnas.2413442122.
- 9 **N. N. Chaudhari**, P. E. Imms, N. F. Chowdhury, *et al.*, "Increases in regional brain volume across two native south american male populations," *GeroScience*, pp. 1–21, 2024.  DOI: 10.1007/s11357-024-01168-2.
- 10 K. H. Guo, **N. N. Chaudhari**, T. Jafar, *et al.*, "Anatomic interpretability in neuroimage deep learning: Saliency approaches for typical aging and traumatic brain injury," *Neuroinformatics*, pp. 1–16, 2024.  DOI: 10.1007/s12021-024-09694-2.
- 11 P. Imms, **N. N. Chaudhari**, N. F. Chowdhury, *et al.*, "Neuroanatomical and clinical factors predicting future cognitive impairment," *GeroScience*, pp. 1–20, 2024.  DOI: 10.1007/s11357-024-01310-0.
- 12 P. Imms, N. F. Chowdhury, **N. N. Chaudhari**, *et al.*, "Prediction of cognitive outcome after mild traumatic brain injury from acute measures of communication within brain networks," *Cortex*, vol. 171, pp. 397–412, 2024, ISSN: 0010-9452.  DOI: 10.1016/j.cortex.2023.10.022.
- 13 R. J. Massett, A. S. Maher, P. E. Imms, *et al.*, "Regional neuroanatomic effects on brain age inferred using magnetic resonance imaging and ridge regression," *The Journals of Gerontology: Series A*, vol. 78, no. 6, pp. 872–881, 2023.  DOI: 10.1093/gerona/glac209.
- 14 C. Yin, P. Imms, M. Cheng, *et al.*, "Anatomically interpretable deep learning of brain age captures domain-specific cognitive impairment," *Proceedings of the National Academy of Sciences*, vol. 120, no. 2, e2214634120, 2023.  DOI: 10.1073/pnas.2214634120.
- 15 A. Irimia, V. Ngo, **N. N. Chaudhari**, *et al.*, "White matter degradation near cerebral microbleeds is associated with cognitive change after mild traumatic brain injury," *Neurobiology of aging*, vol. 120, pp. 68–80, 2022.  DOI: 10.1016/j.neurobiolaging.2022.08.010.
- 16 D. J. Robles, A. Dharani, K. A. Rostowsky, *et al.*, "Older age, male sex, and cerebral microbleeds predict white matter loss after traumatic brain injury," *Geroscience*, vol. 44, no. 1, pp. 83–102, 2022.  DOI: 10.1007/s11357-021-00459-2.
- 17 A. Irimia, **N. N. Chaudhari**, D. J. Robles, *et al.*, "The Indigenous south american tsimane exhibit relatively modest decrease in brain volume with age despite high systemic inflammation," *The Journals of Gerontology: Series A*, vol. 76, no. 12, pp. 2147–2155, May 2021, ISSN: 1079-5006.  DOI: 10.1093/gerona/glab138. eprint: <https://academic.oup.com/biomedgerontology/article-pdf/76/12/2147/41150737/glab138.pdf>.
- 18 A. Irimia, A. S. Maher, **N. N. Chaudhari**, N. F. Chowdhury, E. B. Jacobs, and A. D. N. Initiative, "Acute cognitive deficits after traumatic brain injury predict alzheimer's disease-like degradation of the human default mode network," *Geroscience*, vol. 42, no. 5, pp. 1411–1429, 2020.  DOI: 10.1007/s11357-020-00245-6.
- 19 **N. N. Chaudhari**, A. Nagpal, and H. Santhi, "Bpm development for insurance claims using pega," *International Journal of u-and e-Service, Science and Technology*, vol. 10, no. 7, pp. 91–106, 2017.  URL: https://article.nadiapub.com/IJUNESST/vol10_no7/9.pdf.
- 20 **N. N. Chaudhari**, A. Sharma, and A. M. Lal, "Crack detection using image processing for automobiles and aircrafts," *International Journal of Engineering Technology Automobile Security*, vol. 1, no. 1, pp. 17–28, 2017.  DOI: 10.21742/ijetas.2017.1.1.03.
- 21 P. Mani, A. Thangavelu, A. Sharma, and **N. N. Chaudhari**, "A real time monitoring system for yoga practitioners," *International Journal of Intelligent Engineering & Systems*, vol. 10, no. 3, pp. 83–93, 2017.  DOI: 10.22266/ijies2017.0630.10.
- 22 A. Baskaran, **N. N. Chaudhari**, R. D. Caytiles, and N. C. S. Iyengar, "Fuzzy optimized and bee inspired routing protocol for improved qos in mobile ad hoc networks," *International Journal of Control and Automation*, vol. 9, no. 8, pp. 391–402, 2016.  DOI: 10.14257/ijca.2016.9.8.37.
- 23 **N. N. Chaudhari**, A. Gupta, and S. Raju, "Aled system to provide mobile iot assistance for elderly and disabled," *International Journal of Smart Home*, vol. 10, no. 8, pp. 35–50, 2016.  DOI: 10.14257/ijsh.2016.10.8.05.

- 24 J. Jayashree, **N. N. Chaudhari**, and N. C. S. Iyengar, "Promethean fuzzy model to predict diabetic foot ulcer," *International Journal of Hybrid Information Technology*, vol. 9, no. 11, pp. 181–198, 2016.  DOI: 10.14257/ijhit.2016.9.11.16.

Conference Proceedings

- 1 **N. N. Chaudhari**, N. F. Chowdhury, E. Dennis, E. Wilde, and A. Irimia, "Synthesizing brain magnetic resonance images for myelin assessment," in *2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, IEEE, 2026, pp. 1–5.  URL: <https://drive.google.com/file/d/1AgjAMP4FaLzWH2OQjyfIBVBov6qXbbzD/view>.
- 2 P. Imms, **N. N. Chaudhari**, O. M. Vega, and A. Irimia, "Local brain aging maps proximity to alzheimer's-related cognitive impairment," in *Alzheimer's Association International Conference*, Alzheimer's Association, 2025.
- 3 G. Liu*, **N. N. Chaudhari***, N. Kanakaris, C. Yin, P. Bogdan, and A. Irimia, "Controllable generative model for brain evolution," in *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, *Equal contribution, 2025, pp. 1–5.  DOI: 10.1109/ICASSP49660.2025.10888742.
- 4 C. Allen, **N. N. Chaudhari**, and A. Irimia, "Neuroimaging data augmentation for robust brain age estimation using deep neural networks," in *21st IEEE Conference on Computational Intelligence in Bioinformatics and Computational Biology (CIBCB)*, IEEE, 2024.  URL: https://cmte.ieee.org/cis-bbtc/wp-content/uploads/sites/172/IEEE_CIBCB_2024_paper_82.pdf.
- 5 **N. N. Chaudhari**, P. E. Imms, N. F. Chowdhury, *et al.*, "Cross sectional sex differences in cortical atrophy across an indigenous bolivian population," Organization of Human Brain Mapping (OHBM), 2024.
- 6 H. Kaplan, J. Aronoff, P. Hooper, *et al.*, "Allometry of the brain, brain regions and body size among tsimane and moseten native south americans," in *American Journal of Human Biology*, Wiley 111 River St, Hoboken 07030-5774, NJ USA, vol. 36, 2024.
- 7 J. Kawamura, **N. N. Chaudhari**, N. F. Chowdhury, P. Imms, and A. Irimia, "Interpretable deep learning reveals differences in brain aging according to genetic risk of alzheimer's disease.," Biomedical Engineering Society, 2024.
- 8 **N. N. Chaudhari**, A. S. Maher, P. E. Imms, *et al.*, "Significantly different relationships of basal ganglia volumes with age in south american tsimane compared to cognitively normal us adults," in *Alzheimer's Association International Conference*, ALZ, 2022.
- 9 A. Irimia, N. Chowdhury, S. Wang, *et al.*, "Widespread cortical demyelination in geriatric cases of mild traumatic brain injury and in alzheimer's disease," in *Innovation In Aging*, Oxford Univ Press Great Clarendon St, Oxford OX2 6DP, England, vol. 5, 2021, pp. 673–674.
- 10 D. Robles, A. Dharani, K. Rostowsky, *et al.*, "Chronic white matter degradation after mild traumatic brain injury as a function of age, sex and cerebral microbleed count," in *Journal of Neurotrauma*, Mary Ann Liebert, Inc 140 Huguenot Street, 3rd Fl, New Rochelle, NY 10801 USA, vol. 38, 2021, A21–A21.
- 11 A. Irimia, D. Fan, **N. N. Chaudhari**, *et al.*, "Mapping cerebral connectivity changes after mild traumatic brain injury in older adults using diffusion tensor imaging and riemannian matching of elastic curves," in *2020 IEEE 17th international symposium on biomedical imaging*, IEEE, 2020, pp. 1690–1693.
- 12 A. Maher, E. Jacobs, **N. N. Chaudhari**, and A. Irimia, "Default mode network differences between geriatric mild traumatic brain injury and alzheimer's disease," in *Journal of Head Trauma Rehabilitation*, Lippincott Williams & Wilkins Two Commerce Sq, 2001 Market St, Philadelphia, USA, vol. 35, 2020, E227–E227.
- 13 D. Fan, **N. N. Chaudhari**, K. A. Rostowsky, *et al.*, "Post-traumatic cerebral microhemorrhages and their effects upon white matter connectivity in the aging human brain," in *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, IEEE, 2019, pp. 198–203.
- 14 D. Fan, **N. N. Chaudhari**, K. A. Rostowsky, *et al.*, "The effects of cerebral microhemorrhages on white matter connectivity and aging in mild traumatic brain injury," in *Journal of Neurotrauma*, Mary Ann Liebert, Inc 140 Huguenot Street, 3rd Fl, New Rochelle, NY 10801 USA, vol. 36, 2019, A129–A129.

- 15 A. Irimia, K. A. Rostowsky, **N. N. Chaudhari**, M. Calvillo, S. K. Lee, and A. Lenartowicz, "Mild traumatic brain injury in older adults and alzheimer's disease share patterns of white matter degeneration," in *Journal of Neurotrauma*, Mary Ann Liebert, Inc 140 Huguenot Street, 3rd Fl, New Rochelle, NY 10801 USA, vol. 36, 2019, A129–A129.

Book Chapter

- 1 **N. N. Chaudhari**, R. S. Bhadoria, and S. Prasad, *Information handling and processing using enterprise service bus in service-oriented architecture system*, IEEE, 2016.  DOI: 10.1109/CICN.2016.88.

Abstracts

- 1 P. Imms, **N. N. Chaudhari**, N. Chowdhury, *et al.*, *Two native south american male populations exhibit cross-sectional increase in regional brain volume with age*, 2023.  DOI: 10.1093/geroni/igad104.3676.
- 2 P. Imms, J. Kawamura, C. Yin, *et al.*, *Early identification of alzheimer's risk using anatomically interpretable estimation of brain age*, 2023.  DOI: 10.1002/alz.082676.
- 3 N. Kim, A. Amgalan, P. Imms, N. Chowdhury, **N. N. Chaudhari**, and A. Irimia, *Mutual information of neuroimages and genetic variants to study the determinants of brain aging*, 2023.  DOI: 10.1093/geroni/igad104.3483.
- 4 N. Kim, P. Imms, **N. N. Chaudhari**, *et al.*, *Reproducibility of brain age saliencies across deep neural network architectures*, 2023.  URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10737381/pdf/igad104.3572.pdf>.
- 5 A. Irimia, A. Maher, K. Rostowsky, *et al.*, *Early prediction of cognitive deficits after traumatic brain injury based on ad-like patterns of neurodegeneration*, 2021.  DOI: 10.1093/geroni/igab046.1439.
- 6 D. Robles, A. Dharani, **N. N. Chaudhari**, *et al.*, *Age, sex, and cerebral microbleeds affect white matter integrity across adulthood after mild traumatic brain injury*, 2021.  DOI: 10.1093/geroni/igab046.3021.
- 7 A. Irimia, K. Rostowsky, **N. N. Chaudhari**, M. Calvillo, and S. Lee, *Geriatric traumatic brain injury and alzheimer's disease share similar patterns of white matter degradation*, 2019.  DOI: 10.1093/geroni/igz038.363.
- 8 K. A. Rostowsky, **N. N. Chaudhari**, M. Calvillo, S. K. Lee, and A. Irimia, *Geriatric traumatic brain injury and alzheimer's disease share patterns of white matter decline*, 2019.  DOI: 10.1016/j.jalz.2019.06.3975.

Awards, Grants and Patents

- 2026  **GeroScience Publication Award Honorable Mention** *The American Aging Association*
Honorable mention at American Aging Association 54rd Annual meeting for Increases in regional brain volume across two native South American male populations [9].
- 2024-26  **Honorarium** *CURVE Mentor at Viterbi School of Engineering, USC*
- 2025  **US Patent** A Irimia & **N. N. Chaudhari** *Regional segmentation and parcellation of the human cerebral cortex from computed tomography*. Stevens Center for Innovation, University of Southern California. Disclosed on March 6, 2023 under application no. 2023-079. Provisional patent application 63/530,881 filed with USPTO on August 4, 2023 under submission no. 71523-43150. Application for conversion to non-provisional patent filed on August 5, 2024 under US patent application 18/794,766.
-  **Scholarship Award** *India Conference at USC*
Conference organized by USC Marshall school of Business and the South Asian Business Association (SABA) to advance the South Asian community academically and professionally.

Awards, Grants and Patents (continued)

- 2024  **Resource Award** *National Artificial Intelligence Research Resource*
Generative models of neuroimages for personalized assessment of risk for Alzheimer's disease. 500 Node hours on TACC Vista with a NVIDIA GH100 Grace Hopper Superchip GPU. Award ID NAIRR240132.

Mentoring and Leadership

- 2024, 2025  **IUSSTF-Viterbi mentor** Mentored undergraduate students during summer through the Indo-US Science and Technology Forum – Viterbi program.
- Fall 2021,
2024 – 2025  **CURVE mentor** Mentored undergraduate students through Center for Undergraduate Research in Viterbi Engineering (CURVE) assisting them to explore research areas. Trained the students in technical knowledge and interpersonal skills required to become successful researchers.
- 2018 – 2019,
2021 – 2022, Fall 2024  **Viterbi Graduate mentor** Mentored new graduate students during their transition to USC.
- Hindu Student Organization*
- 2018- 2019  **Program Manager** Organized events including Diwali and Janmashtami for the Indian community at USC.
- 2017- 2018  **Volunteer** Organized events including Janmashtami and temple trips.

Additional academic work

- 2026  *Nature Progress Brain Health*
Jornal Article Reviewed An age-adaptive multimodal biomarker panel for mixed-pathology brain health.
-  *npj Dementia*
Jornal Article Reviewed Investigating Individual Differences in Brain Structure for Brain Disorder Diagnosis: Insights from Aging, Genetics, and Modifiable Factors.
-  *Imaging Neuroscience*
Jornal Article Reviewed Heterogeneous Brain Aging of Cerebral Networks in Subjective Cognitive Decline.
- 2025  *Brain Structure and Function*
Jornal Article Reviewed Occupational Circadian Disruption and Brain Structure: Evidence from Normative Developmental Brain Lifespan Trajectories.

Additional academic work (continued)

IEEE International Symposium on Biomedical Imaging (ISBI) 2026
Conference Articles Reviewed

1. RadGaze-LLM: Anatomical Region-Grounded Radiology Report Generation via Learning from Expert Gaze (Paper number 1571222455).
2. Mitigating Affirmation Bias in Vision-Language Models for Medical Image Segmentation (Paper number 1571220336).
3. Structure-Dynamics-Informed ICA Model for Functional Brain State Estimation (Paper number 1571226515).

IEEE Transactions on Medical Imaging
Journal Article Reviewed Brain Age Prediction via Adversarial Training and Multi-Model Evaluation: A Region-Specific Analysis of Aging Differences in Parkinson’s Disease (Manuscript Number TMI-2025-1273).

Neuroimage: Clinical
Journal Article Reviewed Associations of traumatic brain injury with markers of brain health among older men in the Vietnam Era Twin Study of Aging (Manuscript Number NICL-24-799).

2024 Neuroimage
Journal Article Reviewed Age- and Sex-specific Normative Curves for Intracranial and Ventricular Volumes from Head CT Scans (Manuscript Number NIMG-24-1486).

USC Viterbi School of Engineering

2023 **Finalist** in the three-minute thesis (3MT) competition.

2022 Presentation on basics of Linux terminal.

ENIGMA-U (a free online school focused on neuroscience) Presentation about basics of human brain structure and aging.

Professional Memberships

2025 - IEEE Signal Processing Society
IEEE Young Professionals.

Skills

Research & ML Core	PyTorch, TensorFlow, diffusion models, ControlNet, generative modeling, DNN architecture design, large-scale training on NVIDIA A100 GPUs
Research Methodology	Independent research agenda development, problem formulation, hypothesis testing, experimental design, statistical analysis (regression, classification), first-author publication track record
ML Engineering	GPU programming (CUDA), performance optimization, memory-efficient inference, distributed training, end-to-end pipeline development

Skills (continued)

Interpretability & Reproducibility	■ xAI (integrated gradients, SHAP), explainable model design, open-source tool development
Cloud & Infrastructure	■ AWS (EC2, S3), Google Cloud (GCP), GPU cluster administration, cost-aware resource optimization
Software Engineering	■ Python, C++, Java, MATLAB, Linux/bash, OOP, GitHub actions, CI/CD, production-ready code development
Data & Tools	■ pandas/numpy, ETL pipelines, Elasticsearch, Kibana, SQL, JSON, Oracle 11g
Model Deployment & Optimization	■ ONNX export/validation, ONNX Runtime (onnxruntime), TensorFlow/PyTorch-to-ONNX conversion
Interactive & Web Applications	■ Gradio, Streamlit, Plotly, HTML5 Canvas 2D rendering, JavaScript/ES modules, REST/API integration, client-server preprocessing
DevOps & Containerization	■ Docker, Docker Compose orchestration, CI/CD-ready pipelines, GPU-accelerated container workloads
Design	■ Adobe Creative Suite (Photoshop, Illustrator, Audition).
Collaboration & Communication	■ Cross-functional research collaboration, project ownership and autonomy, translating complex research into practical solutions